Decision memo — NVIDIA Corporation (NASDAQ: NVDA)

To: the investment committee (no current position) · **From:** [student name], AI task-force analyst, sector group · **Valuation date:** 23 Sep 2026 (close $225.51) · **Information cutoff:** Q2 FY2027 10-Q, filed 26 Aug 2026 · **Currency / share basis:** USD; 24,195m diluted shares (24,100m outstanding at 21 Aug 2026 + 95m dilutive awards) · **Repository:** github.com/Jesstran12/NVDIA-Research (commit [fill in at freeze]) · *Learning exercise for FIN 43900; not investment advice.*

1. Decision and committee ask

Do not initiate at $225.51; keep NVIDIA on watch. The FCFF DCF driven by my five-year pro-forma gives **$124 per share in the base case and $80–$187 across the causal low and high cases**; the price sits above the high case. The market is paying for about **24 points a year more revenue growth in FY2028–FY2031** than the filings and guidance support (a path near 49 / 36 / 31 / 28% instead of 25 / 12 / 7 / 4%), or for a cost of equity near 8% on a stock whose measured beta is above 1.8. Neither is in the evidence yet; both are testable within two quarterly reports.

Ask: approve the watch posture with no position; authorise a re-review within one week of the Q3 FY2027 report (November 2026), with this tool re-run on the new filing. **Initiate** if the price falls below the base case with the checks still passing, or if two quarters of Data Center growth and margin evidence lift the high case above the price.

2. Evidence and as-of date

Evidence	Value	Source
FY2026 revenue / operating income / net income	215,938 / 130,387 / 120,067	10-K FY2026, income statement p. 51 (accession 0001045810-26-000021)
H1 FY2027 revenue (+96% y/y) / operating income	177,837 / 117,270	10-Q Q2 FY2027, p. 3 (accession 0001045810-26-000075)
Q3 FY2027 outlook	revenue $108.0bn ±2%; gross margin 74.0%; opex ≈ $9.2bn; FY tax 16–18%	Q2 FY2027 release, 26 Aug 2026
Customer concentration	two direct customers = 36% of FY2026 revenue	10-K MD&A p. 41
Opening balance sheet (bridge)	cash 10,605; marketable debt 39,065; marketable equity 12,886; debt 8,468	10-K p. 53; 10-Q comparative column
Supply and purchase commitments	$279bn, $92bn due in the rest of FY2027	10-Q Note 10
Measured beta	2.21 raw (s.e. 0.30), 1.81 Blume; CAPM 13.0–14.8%	`beta.py`, 60 monthly returns to Sep 2026

Facts are from the filings; FY2027E is guidance plus one quarter of judgment; everything after FY2027 is my assumption, labelled as such in the model. Every input, locator and check is in `docs/evidence-ledger.csv`.

3. Valuation (USD millions; FCFF, five explicit years, Gordon terminal; checks all pass)

	FY2027E	FY2028E	FY2029E	FY2030E	FY2031E
Revenue (growth 85.6 / 25 / 12 / 7 / 4%)	400,781	500,976	561,093	600,370	624,385
Operating income (gross margin 74.5 → 70%; opex 8.7 → 12.5%)	263,714	318,120	345,072	357,220	359,021
FCFF = EBIT(1 – 17%) + D&A + SBC – capex – ΔWC	201,158	256,256	286,205	300,892	305,970

PV of explicit FCFF 952,816 + PV of terminal value 1,991,557 (3% growth; 67.6% of EV) = **enterprise value 2,944,373**; + cash and marketable securities 62,556 − debt 8,468 = **equity 2,998,461**; ÷ 24,195m = **$123.93**. WACC 11.99% (cost of equity 12%, debt weight 0.16%). The Lab 10 FCFE cross-check gives $123.05. Balance sheet balances, cash ties, base year reconciles to the 10-K, bridge reconciles, value falls monotonically as WACC rises.

Cases (full reruns, never sums of shocks): low $80 (growth path 15 / 2 / –3 / –6, gross margin –3 pp, opex +1 pp, 13% cost of equity: capex digestion plus custom-silicon share loss) · base $124 · high $187 (growth 35 / 22 / 17 / 14, gross margin +1.5 pp, 11%: the cycle compounds and CUDA holds price).

Cross-checks that disagree, not averaged: NVIDIA trades at 28.5x trailing GAAP earnings and 27.4x trailing EBIT; Broadcom's 45.3x / 40.4x applied to NVIDIA imply $358 / $332; AMD's 158x / 153x are reported but not meaningful under the pre-declared rule (its 16% operating margin is a recovery being capitalised). Multiples capitalise last year's earnings at the growth the market expects; the DCF capitalises the growth the filings support. The gap is the disagreement, and the reverse DCF puts it in operating language: +24 pp growth a year, or 8% cost of equity, or 7.8% terminal growth; no margin or cost assumption inside the defended bounds reaches $225.

4. Material assumptions and sensitivities

One driver at a time at its defended bounds (base $123.93): growth path FY2028–31 –25/+25 pp → $63 / $233 (span $170); cost of equity 16% / 9% → $86 / $186 ($101); gross margin –13/+4 pp → $97 / $132 ($36); terminal growth 0 / 4% → $102 / $135 ($34); FY2027E revenue ±10% → $112 / $136 ($24); opex, tax, SBC, capex under $17 each. The ranking's top driver is the same at half the bounds and at the Lab 11 ranges; the middle order depends on the bounds. **Locked Changed-Input Record (Lab 11, committed 29 Sep 2026 before the run):** predicted growth would move value about three times as much as margin; result $32.77 vs $12.57 (2.6×), direction and order right, size of the growth effect smaller than predicted (–12.6% vs –15 to –20%). Full record in `docs/validation-and-ai-use.md`.

5. Principal risks, failure conditions and monitoring triggers

Trigger (metric · threshold · window)	Effect on the case
Data Center revenue falls quarter on quarter · any decline · next two reports (Nov 2026, Feb 2027)	withdraw the base; rebuild as the low case (about $80)
Gross-margin guidance below 72% and falling · next two reports	gross-margin path –3 pp (about –$12)
Operating expenses grow faster than revenue · two consecutive quarters	opex path +2 pp (about –$5)
Bridge fails to reconcile after the next 10-Q (debt, buybacks, securities) · each filing	reject the per-share conclusion until it reconciles
Price falls below $124 with checks passing · any day	re-open the initiation question

Known limitations: values struck at the opening balance-sheet date and compared with a price eight months later (rolling forward adds about 8% before distributions); FY2027 is a 53-week year; SBC is added back (cash-cost alternative –$6); the terminal year reinvests only 2.5% of net income while growing 3%.

6. Action requested and next review point

Approve **watch / defer — no position at $225.51**. Monitoring owner: the analyst, re-running this tool on each filing; the sector analysts own the demand view. Next review: within one week of the Q3 FY2027 report. AI assistance and every material check are recorded once, in `docs/validation-and-ai-use.md`.